BOSTON ARCHITECTURAL COLLEGE
SINCE 1889

DME2100 - Digital Narratives

SYLLABUS

Instructor: Hamze Machmouchi
Year and term: FALL 2026
Meeting time: Monday 4 pm EST
Location: Online via zoom
Credits: 3 credits

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Office Hours: 4PM EST – 6PM EST

1. COURSE INFORMATION

COURSE DESCRIPTION:
This course explores spatial narratives in the unbuilt world. From games, movies, AR/VR, to metaverse, students will explore the idea of space in the digital realm. The discussion will be on: How does one interact with space? And how does that interaction with space shape a spatial narrative?

This workshop encompasses both theoretical readings on spatial narratives as well as learning and working with three-dimensional software in order to compose a holistic perspective of three-dimensional narratives in the virtual realm. Students will be required to play virtual games, study their spaces and narratives, and create digital narratives.

This semester the game analysis is expanded. Students take their assigned game apart in four layers, starting with its design architecture and moving to its mechanics, and document it in six diagrams. Rhino and Unreal begin in Week 1 and run alongside the analysis. Each student is also assigned a medium, which governs both the drawings and the camera setup in Unreal for the rest of the semester. The second half of the semester is spent building an interactive world, in which an original game mechanic is one of the instruments used to tell the story.

The primary reference is *An Architectural Approach to Level Design* by Christopher Totten, supported by readings and tutorials provided throughout the semester.

LEARNING GOALS

1. Cultivating an understanding of the history and theory of virtual design.
2. Learning mesh-modeling and SubD in Rhino 8, and implementing, animating, and filming in Unreal Engine 5.
3. Developing spatial narratives and exploring user interfaces through gamification.
4. Reading a game space as built work through diagrams.
5. Designing an original game mechanic that carries a narrative.
6. Incorporating A.I. tools into the Rhino and Unreal workflow.

COURSE STRUCTURE
This course meets every Monday at 4 pm EST via zoom, for thirteen sessions between August 24 and November 30. One is a studio session held on Discord, where the instructor is available throughout the scheduled class time. Two further class dates fall on BAC closures. Here below are the important dates for the course.

August 24 – First day of classes
September 7 – BAC closed, Labor Day
October 12 – BAC closed, Indigenous Peoples' Day
October 19 – Midterm presentations
November 6 – Withdrawal deadline
November 30 – Final presentations and last day of classes
December 5 – Main session courses end

COURSE EXPECTATIONS
Students are expected to be actively engaged throughout the course. Because the course meets online and only once per week, the following expectations govern the work carried out between sessions. Each contributes to the participation portion of the final grade.

1. **Play is research.** Students are required to play their assigned games and study their spaces and narratives. Take notes and capture screenshots while you play. This material feeds directly into your own project.
2. **The camera follows the medium.** Your assigned medium sets your camera for the semester. Axonometric means orthographic and locked. Section means the model is cut and the camera sits in the cut.
3. **Weekly progress.** Upload progress work to the class drive every week, finished or not. Discord is used for informal communication and technical support between sessions.
4. **Crit partners.** Students are paired for the full semester. Crit partners review one another's work ahead of each review and report back what story they read from it.

WHERE WORK IS POSTED
Discord will be used as a platform for informal communication beyond class sessions. A One Drive dedicated to this class will be used to share readings, resources, and materials related to coursework. Students are expected and encouraged to upload their progress work to this one drive throughout the semester. The course schedule, assignment briefs, and links are kept at https://machmouchih.github.io/digital-narratives-schedule/.

ASSIGNED GAMES
The class is divided into four research groups. Each group analyzes its assigned games, focusing on narrative, level design, and world-building, and presents the analysis in the third session. Groups and games are assigned in the first session, so students should wait until after that session to buy anything. Several of these titles can be streamed to a laptop rather than bought.

- **Group 1, The World Weavers:** Star Wars Jedi: Survivor, Cyberpunk 2077
- **Group 2, The Mind Benders:** Alan Wake 2, Control
- **Group 3, The Intimate Storytellers:** What Remains of Edith Finch, Stray
- **Group 4, The Historical Architects:** A Plague Tale: Requiem, Assassin's Creed Shadows

FINAL PROJECT
The final project consists of each student creating their own digital narrative and 3-dimensional game space in Unreal Engine 5, developed across four phases. The analysis is the opening assignment and is presented in the third session. The eight-second loop is presented at the midterm, and the mechanic and the world are built in the second half.

1. **Game Space Analysis.** The opening assignment, running alongside the Rhino and Unreal workshops rather than before them. Play your group's game, take it apart in four layers, and present the analysis in the third session. You are documenting how the game is built rather than reviewing it, and the analysis is presented in drawings. The four layers are design architecture, or how the world is structured; spatial grammar, or how the space instructs without instructions; game mechanics, or what the player can do and what it means; and pacing, or how the experience rises and falls. Six diagrams are required: a level graph, an annotated plan of one level, a section through a significant vertical moment, a sightline diagram of a single reveal, a core loop diagram, and an intensity curve across thirty minutes of play. This presentation is the Demo portion of the grade.
2. **Verb, Part, and Medium.** Each student is assigned one Architectural Verb and one Medium, and chooses one Building Part as their primary tool for storytelling. The verbs are framing a view, guiding movement, creating thresholds, compressing and releasing, ascending and descending, sheltering and exposing, revealing and concealing, scaling and dominating, grounding and floating, and reflecting and distorting. The parts are wall or façade, floor or ground plane, ceiling or roof, door or opening, window or aperture, stair or ramp, column or structure, and pathway or corridor. The media are section, plan, axonometric, elevation, worm's-eye or reflected ceiling plan, exploded assembly, cutaway diorama, one-point perspective, collage or photomontage, panel grid, and text and typography. The medium sets the drawings produced and the camera built in Unreal Engine. The result is presented at the midterm as an eight-second loop.
3. **Your game mechanic.** In the second half of the semester each student designs a single core mechanic, specifies it, and implements it in Blueprints. The mechanic is not the point of the project; it is the instrument through which the story is told, and what is assessed is the story it tells rather than its technical complexity. The specification is one page, written before any Blueprint work begins, and covers the verb, or what the player can do that they could not do before; the input, or how it is triggered and how the player learns that without being told; the feedback, or what changes in the world so the player knows it worked; the limits, or what stops it and what the limit means; and the meaning, or what the mechanic says that a cutscene could not. Build one mechanic and get it working rather than several that are incomplete. Crit partners review one another's work before the final. Prior programming experience is not required.
4. **The interactive world.** The mechanic is placed in a world built out in Unreal Engine 5, using dynamic materials, advanced lighting, atmosphere, and sound. Students document their process and present the final project in a clear and compelling way, filmed through their assigned medium's camera and packaged for the final review.

PROJECT CRITERIA:

- **Lighting:** how does the quality of light shape the spatial experience?
- **Materiality:** how do different materials shape the digital narrative?
- **Interaction with space and movement:** how does one interact with the space?
- **Narrative:** what is the essential narrative that this experience delivers?

ATTENDANCE
Students are expected to be present during class. Absences accruing more than three times during the semester will result in a reduction of grades at the discretion of the instructor. If there are any extenuating circumstances, please reach out to the instructor to inform them of your situation and to schedule extensions if permitted. Lateness to class of more than 15 minutes will be counted as an absence unless otherwise communicated to the instructor in advance.

EVALUATION CRITERIA
Demo: 10%
Assignments: 15%
Attendance and Participation: 25%
Mid Review: 25%

Final Review: 25%
Total of 100%

ASSISTANCE:
Tutorials will be provided throughout the course in addition to technical support on Discord. Rhino 8 and Unreal Engine 5 should both be installed during the first week, and any problems reported on Discord early rather than the night before a review.

BAC GRADE DEFINITIONS
Students must maintain minimum GPAs, determined by their program, to maintain Satisfactory Educational Progress. Failure to maintain SEP may result in assignment of additional work, repeating a course or semester, or withdrawal from the program. More information, including the GPA required for Satisfactory Educational Progress by program, can be found at Satisfactory Educational Progress | The BAC (the-bac.edu).

BOSTON ARCHITECTURAL COLLEGE OFFICIAL INSTITUTIONAL NAMES: BOSTON ARCHITECTURAL CENTER PRIOR TO JUNE 1, 2006			
GRADE	**4.0 SCALE**	**0 – 100 SCALE**	**DEFINITION**
A	4.0	94 – 100	Excellent. The work exceeds the requirements of the course and demonstrates complete understanding of course goals. In addition, assignments exhibit a level of critical thinking that has allowed the student to demonstrate creative problem solving. Ideas and solutions are communicated clearly, showing a high level of attention and care.
A–	3.7	90 – 93	
B+	3.3	87 – 89	
B	3.0	84 – 86	Good. The work meets the requirements of the course and demonstrates understanding of course goals. The assignments reflect an ability to solve problems creatively, but solutions demonstrate inconsistent depth and critical thinking ability. Ideas and solutions are communicated effectively, but may lack the clarity and depth one sees in excellent work.
B–	2.7	80 – 83	
C+	2.3	77 – 79	
C	2.0	74 – 76	Satisfactory. The work meets the minimum requirements of the course and reflects understanding of some course goals but is lackluster. The assignments exhibit a basic problem-solving ability, but the process and solutions lack sufficient depth and demonstrate a need for greater critical thinking. Ideas are communicated ineffectively, showing a lack of attention to detail and a decided lack of clarity or depth.
C–	1.7	70 – 73	
D	1.0	60 – 69	Less than satisfactory. The work barely meets the minimum requirements of the class. Assignments lack depth and display a minimal understanding of course goals. Ideas are presented with little or no detail or elaboration. Course guidelines are often not followed.
RF Repeat/Fail	0.0	0 – 59	Unacceptable or missing work. The work neither satisfies the requirements of the class nor demonstrates understanding of course objectives. The presentation of work is unprofessional and/or incomplete. Overall, the student shows insufficient understanding of the course requirements. Poor attendance or violation of academic integrity policy may also be factors.
NF	0.0	N/A	Failure due to non-attendance
I	N/A	N/A	Incomplete
W	N/A	N/A	Withdrawn
P	N/A	N/A	Pass. Used only in specially designated courses and educational reviews.
NP	N/A	N/A	No Pass
NC	N/A	N/A	No credit. Used if a student replaces a failing grade. Not included in GPA calculation.
NS	N/A	N/A	No show. Awarded only for Educational Reviews if a student registers but does not attend.
T	N/A	N/A	Transfer credit
WV	N/A	N/A	Waiver

2. COURSE POLICIES AND PROCEDURES

DEADLINES POLICY:
Students should complete assignments to the best of their ability and submit them on time. If circumstances require a late submission, the student should contact the instructor before the assignment is due; instructors may consider appropriate accommodation at their discretion. In the event of an emergency (e.g., medical, personal), the student should contact both the instructor and student advisor as soon as possible.

MID-SEMESTER WARNING
Students will receive a progress assessment at mid-semester. To help students succeed, students who do not perform up to expectations will receive a Mid-Semester Warning; a copy of the warning will be kept in the student's file.

WRITING STANDARDS
Writing in this course should meet the standard of accuracy and the clarity of expression expected of design professionals, such as appropriate grammar, correct spelling, and clear and well-organized arguments.

USE OF A.I.
Artificial intelligence tools are part of the pipeline in this course and are taught in four sessions. Their use is treated as a normal part of the working method rather than as an exception to it, and students are not required to avoid them.

They are used to produce assets and to assist with scripting. The diagrams, the mechanic specification, and the narrative are the student's own work, and students should be able to account for every generated asset in their scene. The second session covers previsualization, using image and video generation for moodboards and concept frames. The fourth covers asset generation, including text and image to 3D for props and secondary geometry; generated meshes come in dirty and are cleaned in Rhino, and each student models their own Building Part. The ninth covers drafting, reading, and debugging Blueprint logic with an assistant, which is useful for structure but unreliable on node-level specifics. The tenth covers texture and material generation, video-based motion capture, and generated ambience.

Students should be able to explain what they generated, what they kept, and what they changed. Documentation of A.I. use is submitted at both reviews, generated assets are attributed, and students should expect to be asked to explain any part of their scene during a review.

Work submitted in this course shall be the original creation of its author, and is allowed to contain the work of others so long as there is appropriate attribution. Students who use AI tools for purposes other than those taught in class, including written research, should indicate where and how those tools were used.

Different courses at the BAC could implement different AI policies. It is the student's responsibility to be aware of and comply with expectations for each course.

The language for our AI policy options draws from the University of Delaware, Duke University, and Harvard College.
Guidry, Kevin. "Considerations for Using and Addressing Advanced Automated Tools in Coursework and Assignments". https://ctal.udel.edu/advanced-automated-tools/. Center for Teaching & Assessment of Learning, November 20, 2024.; "Artificial Intelligence Policies: Guidelines and Considerations." https://learninginnovation.duke.edu/ai-and-teaching-at-duke-2/artificialintelligence-policies-in-syllabi-guidelines-and-considerations/. Learning Innovation & Lifetime Education, August 2024.; "AI Guidance and FAQs". https://oue.fas.harvard.edu/ai-guidance. Harvard College Office of Undergraduate Education, 2025.

3. COLLEGE POLICIES AND PROCEDURES

ACADEMIC INTEGRITY
As stated in the Campus Compact the BAC expects intellectual activities to be conducted with honesty and integrity. Work submitted or presented as part of a BAC course:

- Shall be the original creation of its author;
- Is allowed to contain the work of others so long as there is appropriate attribution; and
- Shall not be the result of unauthorized assistance or collaboration.

Failure to adhere to these guidelines is academic dishonesty and may lead to disciplinary action.

PLAGIARISM
Plagiarism is the act of representing someone else's words, images, or ideas as one's own and is a violation of academic honesty. The institution takes such violations seriously. Please see the policy in the Campus Compact. Any suspected cases of academic dishonesty will be addressed according to this policy.

COPYRIGHT COMPLIANCE NOTICE
Courses may contain material used in compliance with the U.S. Copyright Law, including the TEACH Act and principles of "fair use." These materials are made available for the educational purposes of students enrolled at the Boston Architectural College. No further reproduction, transmission, or electronic distribution of this material is permitted. Course materials may not be saved, copied, printed, or distributed without permission other than as specified to complete course assignments. Use of the course materials is limited to enrolled class members for the duration of the course only.

GRIEVANCES
Students may bring grievances to the attention of any academic advisor, administrator, coordinator, or director, who will notify the Dean of Students. The student will be informed of grievance procedures, as well as the degree to which confidentiality may be maintained. The student will be kept informed of the proceedings and given an approximate schedule for investigation and resolution. Please refer to the Campus Compact for more information.

DIVERSITY STATEMENT
The Boston Architectural College is committed to promoting a community that celebrates, affirms, and vigorously pursues inclusiveness in all its forms. (For the full text, refer to the Campus Compact).

DISABILITY SERVICES
The Boston Architectural College complies with the Americans with Disabilities Act and Section 504 of the Rehabilitation Act. The BAC offers reasonable accommodations to students who otherwise cannot reach their academic potential due to a learning disability, physical impairment, medical/psychological condition, or unforeseen circumstances that may arise during the course of their studies. All forms of accommodation are tailored specifically to the individual student and meet guidelines for educational benefit and academic consistency. Accommodations must maintain academic integrity and a realization of required learning objectives. Students who are eligible for accommodations are strongly encouraged to notify the instructor. Students must have appropriate documentation on file with the Disability Services office.

Students seeking accommodation based on a documented disability and/or diagnosis, please contact Disability Services to discuss reasonable accommodations. The Disability Services Director can be reached by emailing DisabilityServices@the-bac.edu. The Disability Services office is located on the first floor of 320 Newbury Street. While you may activate accommodations at any time during your academic career at the BAC, it is highly encouraged to schedule a meeting as soon as possible.

TITLE IX
It is the goal of the Boston Architectural College to promote an educational environment and workplace that is free of sexual harassment. Sexual harassment of employees, faculty or students occurring in the workplace or in other settings in which employees, faculty or students may find themselves in connection with their involvement with the BAC is unlawful and will not be tolerated by this organization. Further, any retaliation against an individual who has complained about sexual harassment or retaliation against individuals for cooperating with an investigation of a sexual harassment complaint is similarly unlawful and will not be tolerated.

We have provided a procedure by which inappropriate conduct will be dealt with, if encountered by employees, faculty, or students in their involvement with the BAC. For the full policy please refer to the Campus Compact.